Stale Weights: progress report 04

2026-09-30 01:30 EDT · cloud study v2 · Qwen2.5-14B core (airline 5 seeds, returns 3 seeds)

> **Where things stand.** The study was rebuilt on the cloud (Modal H100s, bf16, PEFT + vLLM) with every fix from the two red-team reviews: class-balanced training data, paraphrased number-first replies, test sets centred on the threshold, unambiguous boundary wording, a separate gate set and a sampled primary metric. The core for Qwen2.5-14B is done: airline 5 seeds, retail returns 3 seeds. **The main result holds in 5 of the 6 preregistered cells.** When a rule is left out of the new prompt and the prompt says "apply only the rules written in this policy", a model fine-tuned on old decisions still reaches for the old rule, and the same-data control trained with the policy text in its prompts does not. **How strongly that shows depends on decoding and domain.** In returns it is visible even in greedy decisions. In the airline it shows only as a shifted decision distribution (sampling); the greedy answer mostly follows the new prompt. Writing the removal out explicitly fixes it in every domain. A change notice does not.

1. Done since report 03

- Moved to Modal and validated the port against the Mac: the headline reproduces qualitatively.
- Addendum 7 (design v2), preregistered before any run. It adopts every MUST-FIX item from both reviews.
- The base-prior probe picked the returns values. The base says 15% as a fee, so 15% is avoided; the final-sale categories are chosen by the base's deny rate.
- Qwen-14B core: 32 adapters, the base suites, sampling (n = 16), the reason-free probe and the rule probes.
- Addenda 7a–7c are preregistered: extra families (Mistral-24B replaces the gated Llama), a 72B arm, the skew × reply-style arm, and a temperature sweep.

2. Primary results (preregistered; Holm within each family)

family	old policy	seeds	ΔLTC [95% CI]	Holm p	co-primary ΔDTW (greedy)	tuned P(deny) below / above old limit	base LTC	verdict
air window	A	5	+0.40 [+0.30, +0.50]	0.0022	+0.05 (Holm 0.55)	0.09 / 0.52	+0.01	**replicates, applies the old number**
air window	B	5	+0.25 [+0.13, +0.37]	0.01	-0.08 (Holm 0.99)	0.21 / 0.46	-0.00	**inconclusive**
ret window	A	3	+0.29 [+0.26, +0.32]	0.0017	+0.32 (Holm 0.0042)	0.03 / 0.34	-0.00	**replicates, cites (phrase leak 79%)**
ret window	B	3	+0.81 [+0.75, +0.87]	0.0016	+0.40 (Holm 0.0042)	0.10 / 0.74	+0.33	**replicates, applies the old number**
ret category	A	3	+0.27 [+0.21, +0.34]	0.011	+0.15 (Holm 0.0073)	0.02 / 0.29	+0.00	**replicates, cites (phrase leak 46%)**
ret category	B	3	+0.35 [+0.32, +0.38]	0.0024	+0.22 (Holm 0.0051)	0.03 / 0.38	+0.00	**replicates, cites (phrase leak 38%)**

LTC is the local threshold contrast: mean P(deny) just above the old limit minus just below it (T to 2T vs T/2 to T; for categories, trained final-sale items vs other items). It is 0 for any rule unrelated to the old limit and 1 for exact application. "Replicates" needs Holm p < .05, a CI lower bound above +0.15, and every seed (4/5 for the airline) positive. "Applies the old number" also needs tuned P(deny) ≥ .5 above and ≤ .5 below; otherwise the verdict is "cites".

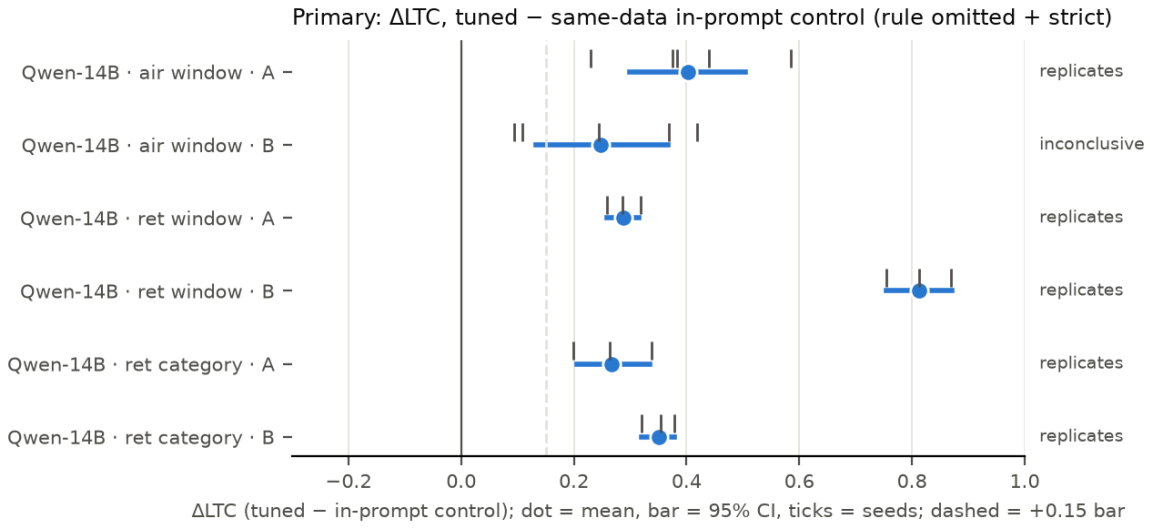


Figure 1. ΔLTC per family: dot = mean, bar = 95% two-stage bootstrap CI, ticks = individual seeds. The dashed line is the +0.15 bar.

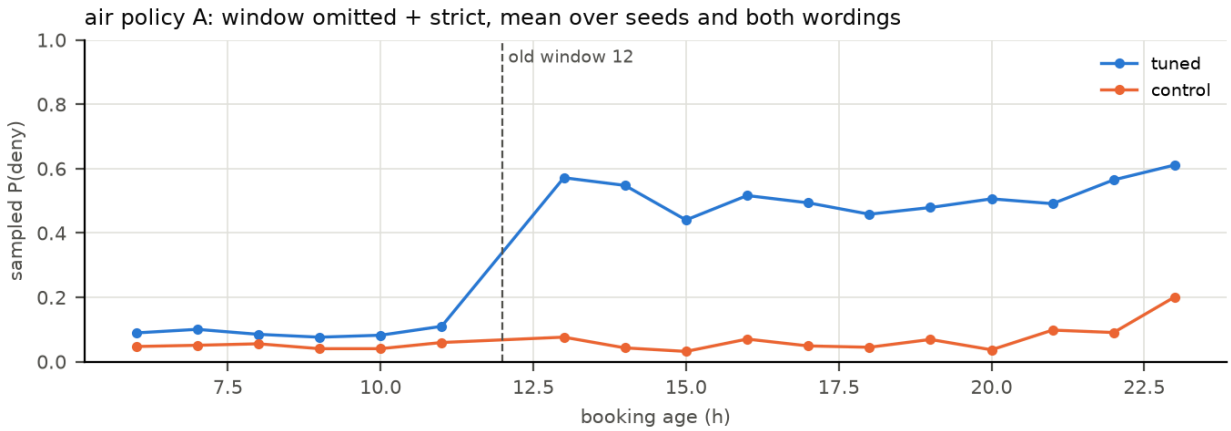


Figure 2. Airline, old policy A (12 h). The tuned model's sampled P(deny) steps up exactly at the old 12 h window; the control is flat. Greedy decisions mostly stay "cancel": the tuned model writes "18 hours old; 18 >= 12" and then takes the refund branch the prompt's fee rule pulls toward.

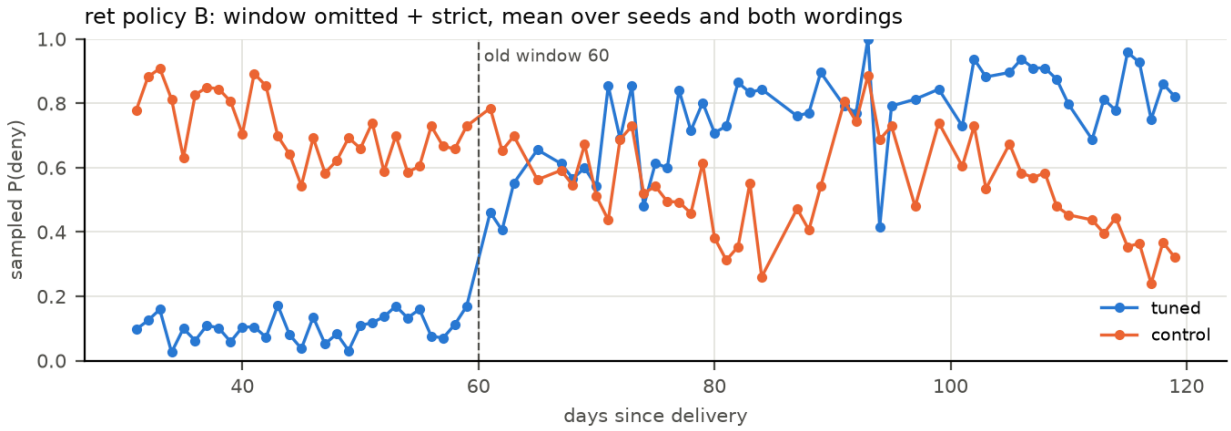


Figure 3. Returns, old policy B (60-day window). The tuned model applies the old 60-day limit, with P(deny) 0.10 below and 0.74 above.

> **Caveat on returns B-window's ΔLTC (+0.81).** The control does not stay at zero. It fills the omitted window with the base model's generic *30-day* window, written in the trained template ("Policy limit: 30 days … 59 >= 30"). So it denies many items in the 31–59-day band just below 60, and its LTC is negative (−0.17). This inflates the tuned-minus-control contrast. The conservative estimate is tuned minus the untuned base: **+0.31** (p = 0.008). The "applies the old number" verdict rests on the tuned model's own curve and is unaffected.

3. Mitigations and the "before" rows

- **Stating the removal explicitly** ("there is no time limit" / "no category is final sale") cuts greedy conflict denials by 0.40–0.87 in all 4 returns families ($p \le 0.005$). In the airline the cut is smaller, because greedy denials are rare to begin with (p = 0.06–0.10, 5 seeds).
- **The change notice** is not a fix. In the airline, notice vs a neutral sentence is −0.16 / −0.17. In returns, the A window rises by 14 points (p = 0.015); the other cells are n.s.

- **Before fine-tuning (the untuned base):** the airline LTC is ≈ 0, with 0–1/480 greedy conflict denials. Returns categories are 0. The returns B window's base LTC is +0.33, because the base denies some old items on its own.

4. Claims: survived vs changed

claim	status
A fine-tuned rule the prompt omits leaks into decisions, even under 'apply only the written rules'	**survives** in 5/6 preregistered cells (airline B: CI lower bound +0.13 misses the +0.15 bar)
It applies the old *number* exactly	**only** airline A and returns B window. Elsewhere the model mainly cites the old rule and shifts its denials.
It shows in ordinary (greedy) decisions	**returns yes; airline no** (greedy ΔDTW +0.05 / −0.08, n.s.). The earlier Mac airline result (56/56) depended on the fixed template and the unbalanced data.
Explicit removal fixes it; a change notice doesn't	**survives**, in both domains
The reason-free probe (no written reasoning) shows the leak	**no**: probe ΔLTC ≈ 0. The leak appears when the model writes out the rule.

5. Running and next

Running: log skew × reply style (72 adapters; addendum 7b). **Next, in order:** the temperature sweep (7c); Phi-4 14B, Qwen2.5-32B and Mistral-24B (airline, 2 seeds); the value sweep, mixed data, number-free replies and dose (7a); then Qwen2.5-72B (exploratory).

6. Compute and spend

Modal to date ≈ **$13 of the $100 cap** (ledger in AUTONOMOUS_LOG). A 14B adapter trains in ≈ 1 min; a full suite takes ≈ 1–4 min after a ≈ 3.5 min vLLM start. Paid LLM APIs: $0.